

Optimization of Aircraft Runway Scheduling Based on Metaheuristic Algorithms

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1. Purpose and Significance of the Study

With the rapid growth of global air transportation, airport congestion has become one of the major bottlenecks limiting the efficiency of air traffic systems. Runway capacity, in particular, plays a critical role in determining airport throughput and overall flight punctuality. Since the construction of new runways is often restricted by high costs, land limitations, and environmental concerns, improving runway utilization through intelligent scheduling has become an important research topic [1][2].

Flight delays not only increase operational costs for airlines but also cause passenger dissatisfaction and environmental pollution due to increased fuel consumption. Statistical studies show that a significant portion of delays originate from inefficient runway scheduling and congestion during peak hours [3]. Therefore, optimizing runway schedules to improve utilization while maintaining safety constraints is of great practical importance.

This study aims to develop an optimization model for aircraft runway scheduling that considers runway availability, flight demand, and operational constraints. The primary objective is to maximize runway utilization, while secondary objectives include minimizing overall flight delays and maintaining scheduling fairness. By applying metaheuristic optimization algorithms to large-scale real-world flight data from Brazil, this research seeks to provide a practical and extensible solution for airport operation optimization.

2. Research Background and Literature Review

The aircraft runway scheduling problem is generally formulated as a combinatorial optimization problem in which aircraft must be assigned to limited runway resources under strict safety constraints. It is widely recognized as an NP-hard problem due to sequence-dependent separation requirements and time window constraints [2].

2.1 Traditional Optimization Approaches

Early research mainly relied on mathematical programming techniques such as mixed-integer linear programming (MILP) and dynamic programming. Beasley et al. proposed a static aircraft landing problem model using mathematical programming, demonstrating its effectiveness on small-scale instances [4]. However, such exact methods suffer from exponential computational complexity and are difficult to apply to large real-world datasets [5].

Bertsimas and Patterson formulated airport scheduling as a flow management problem and highlighted the trade-off between throughput and delay minimization [1]. Later studies confirmed that exact optimization becomes impractical when the number of flights increases significantly [6].

2.2 Metaheuristic Algorithms for Runway Scheduling

To overcome scalability issues, metaheuristic algorithms have been widely adopted. Ant Colony Optimization (ACO) has shown strong performance in handling runway sequencing problems with wake vortex constraints [7]. Genetic Algorithms (GA) [8] and Large Neighborhood Search (LNS) methods have also been successfully applied to improve solution quality within reasonable computation time [9].

Wei et al. proposed a multi-objective optimization model that balances runway utilization and delay minimization, demonstrating that metaheuristic algorithms can efficiently approximate Pareto-optimal solutions [10]. These approaches are particularly suitable for real-world applications where exact solutions are computationally infeasible.

2.3 Reinforcement Learning Approaches

Recently, reinforcement learning (RL) has emerged as an approach for dynamic runway scheduling. Andy et al. applied deep reinforcement learning to runway configuration management and achieved reductions in flight delays [11]. Similarly, Memarzadeh and Kalyanam proposed an offline RL framework that learns optimal runway assignment strategies from historical data [12]. These studies indicate that learning-based methods can adapt well to uncertainty and dynamic traffic patterns.

2.4 Data-Driven Studies and Practical Applications

Large-scale empirical studies based on real flight data have provided valuable insights into delay patterns and congestion mechanisms. Sternberg et al. analyzed Brazilian flight delay data and identified strong correlations between congestion and operational inefficiencies [3]. The Brazilian flight dataset compiled by Teixeira et al. has become a widely used benchmark for flight scheduling research due to its completeness and reliability [13].

3. Research Content, Objectives, and Innovations

3.1 Research Content

This research focuses on building an optimization framework for aircraft runway scheduling based on real-world flight data. The system will:

- Receive flight schedule and operational data as input

- Model runway usage and safety constraints

- Optimize takeoff and landing sequences

- Output an improved runway timetable

The dataset used in this study contains over 2.5 million flight records from Brazilian airports between January 2015 and August 2017, including planned and actual departure times, delay information, airport coordinates, and flight status.

3.2 Optimization Objectives

The primary objective of this research is:

- Maximize runway utilization, defined as the number of successfully scheduled aircraft per unit time.

Secondary objectives include:

- Minimizing total and average flight delay

- Maintaining fairness among flights

- Ensuring safety constraints such as wake vortex separation

These objectives will be incorporated into a multi-objective or weighted optimization function [10][14].

3.3 Innovations

The main innovations of this study include:

1. Problem-oriented modeling of runway utilization optimization: Unlike many studies that focus solely on delay minimization, this work formulates the runway scheduling problem with runway utilization as the primary optimization objective, while simultaneously considering delay and safety constraints. This modeling perspective better reflects real-world airport operational priorities.
2. Application of metaheuristic algorithms to large-scale real-world flight data: The proposed approach applies metaheuristic optimization algorithms to a large Brazilian flight dataset covering more than 2.5 million records. The scale and realism of the dataset enable a more practical evaluation compared with studies based on small or synthetic datasets.
3. Comparative evaluation framework for heuristic and learning-based methods: This study establishes a unified experimental framework to compare traditional metaheuristic algorithms with reinforcement learning-based approaches under the same data and evaluation criteria, providing insights into their respective strengths and limitations.

4. Research Methodology and Plan

4.1 Problem Modeling

The runway scheduling problem will be formulated as a constrained optimization problem. Decision variables represent the assignment and sequencing of flights on runways. Constraints include:

Minimum separation time between consecutive aircraft

Time window constraints

Runway availability

Operational feasibility

The objective function will primarily maximize runway throughput while penalizing excessive delays [2][10].

4.2 Algorithm Design

The research will focus on metaheuristic optimization algorithms, including:

Ant Colony Optimization (ACO)

Genetic Algorithm (GA)

Large Neighborhood Search (LNS)

If time permits, reinforcement learning methods such as Q-learning or Deep Q-Networks will be explored for dynamic scheduling [11][15].

4.3 Experimental Design and Evaluation

Performance will be evaluated using:

Average delay per flight

Total delay

Runway utilization rate

Percentage of delayed flights

Results will be compared with baseline strategies such as First-Come-First-Served (FCFS). Experimental evaluation will be conducted using historical Brazilian flight data [3][13].

4.4 Research Schedule

Algorithm Research				
Dataset Processing				
Backend & Algorithm Developing				
Frontend GUI Developing				
Testing & Evaluation				
Implementation				
Final Report Writing				
	January	February	March	April

5. References

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